Mapping How Video Game Innovations Conquer the Global Market

A research proposal on how new kinds of video games travel from one country to another

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1. Abstract

Think of drinks: coffee, tea, soda, and juice. Each has varieties (e.g. green tea, espresso) and sometimes mixes (an espresso martini mixes coffee and alcohol). Video games work similarly: they have broad genres (action, strategy, adventure, etc.) and sub-genres or tags (shooting, puzzle-solving, role-playing, etc.). Modern games often blend these tags into new hybrid genres. Our study asks: **When a new game genre (tag combination) appears, how does it spread across countries and cultures?** We will collect game metadata (genre tags and developer country) from large databases (e.g. IGDB, Steam). We will identify novel tag combinations and note where and when they first appeared. Then we will trace how developers in other countries adopt those combinations over time. Using network and statistical analysis, we will test factors like geographic distance, shared language, common publishers or platforms, and other links between countries. The expected outcome is a set of maps and models showing the “invisible pathways” through which game design ideas travel worldwide. This will shed light on global innovation networks in the $200+ billion video game industry, with implications for researchers, game makers, and policymakers interested in cultural and technological diffusion.

2. Introduction & Background

The world’s video game industry is enormous (Newzoo projects about $213.9 billion in 2026) and highly interconnected. Games come in categories or genres (like action, strategy, adventure, sports, etc.), similar to how beverages are classified (coffee, tea, soda, juice, etc.). Each genre has subcategories or “tags” – for example, Action games can include shooting, platforming, or fighting sub-genres. Just as a barista might invent a new drink by combining espresso and a cocktail (an espresso martini), modern game designers mix different genre elements to create new hybrid games. For instance, a game might combine action with deck-building mechanics, or a role-playing game with puzzle elements.

This blending of genres means that the traditional labels (action, strategy, etc.) are no longer fixed. Researchers have noted that games vary greatly in whether they fit a single genre or multiple genres. Despite the clear importance of these new “crossover” genres, we know little about how these innovations travel globally. Do they start in one country and then spread to others? Which countries or companies are the first movers? Do game ideas flow more easily between nearby countries, or those that share a language or big publishers?

Several factors could shape the spread of game design trends. In many industries, geographic proximity and cultural similarity matter for innovation diffusion. For example, economic geography studies show that video game firms tend to partner with nearby or culturally-similar firms as the industry matures. Likewise, cultural studies find players often prefer games from their own country, suggesting a domestic bias in taste. But little research has directly traced game genre innovations across developer communities worldwide. We aim to fill that gap by mapping the pathways – geographic, cultural, or industrial – through which new game genres spread.

The problem

Some of these new mixes stay close to home. Others become popular in many countries. We do not yet clearly understand why.

Research questions

Main question: How do new game genre mixes travel across borders, and what helps or slows them down?

1. **Structure:** Which people, studios and groups pass these ideas along, and how are they connected?
2. **Technology:** How do tools such as game engines, app stores, and smartphones make the journey easier?
3. **Culture:** Do local tastes, languages, and habits change how a new mix is received?

Why it matters

Game making is a large creative industry, and many new studios are starting in countries far from the old centres. If we understand how ideas travel, these studios can learn from the paths that worked. Researchers also gain a clearer picture of how creative ideas, not only games, move around the world.

3. Literature Review

What is a genre, and how do genres mix?

Game genres are useful but blurry. Wolf (2001) was among the first to sort games into types. Apperley (2006) pointed out that genres depend on how people play, not only on game rules. Arsenault (2009) showed that genres change over time as makers copy, combine, and improve earlier ideas. More recent work uses many small labels instead of one box. Lee et al. (2014) and Heintz and Law (2015) describe games through several layers of tags. This fits our drink picture: one game can carry several tags at once.

How do new ideas spread?

Rogers (2003) explains that ideas spread through channels, over time, among people who are connected. An idea spreads faster when it is useful, simple, and fits what people already do. Granovetter (1973) showed that loose links between groups are often the ones that carry fresh ideas across. Henderson and Clark (1990) found that many new products come from joining old parts in a new way, which is exactly what a genre mix does. Uzzi et al. (2013) found that the most influential science papers mix mostly familiar ideas with a few unusual ones. Hargadon and Sutton (1997) described "brokers", people who carry ideas from one world into another.

The games industry across borders

Game making has always been international. Consalvo (2006) described how Japanese console games and global companies created a mixed culture. Johns (2006) studied how game production is spread over networks of companies with unequal power. Kerr (2017) shows how games travel through makers, publishers, platforms, and national policies. Tschang (2007) looked at how creative teams balance fresh ideas with careful planning. Wolf (2015) collected stories of games from many regions, and Zackariasson and Wilson (2012) give an overview of the whole industry.

Culture and platforms

Straubhaar (1991) found that audiences often prefer media that feels close to their own culture. Iwabuchi (2002) showed that Japanese pop culture travelled widely partly because it could feel culturally neutral. Huhh (2008) described how Korean internet cafés shaped the way people there play online games. Poell, Nieborg and Duffy (2021) explain how platforms, such as app stores, shape what is made and what people see.

The gap

From our reading so far, most studies look at one country, one company, or one genre at a time. Few follow a single genre mix on its journey through many countries, and fewer still look at structure, technology, and culture together. This study aims to fill that gap.

We borrow Rogers' simple idea of an idea spreading, and we sort the reasons for its journey into three pathways: **structure**, **technology**, and **culture**.

4. Research Methodology

To answer our questions, we will use a data-driven historical analysis of video games. Our approach will have several steps:

- **Data collection:** We will assemble a database of video games with their release year, developer (and developer country), platform, and genre tags. We plan to use sources like IGDB (an open video game database API) and the Steam game API or datasets (e.g. Kaggle Steam data), which include community tags for game mechanics. This will give us a large sample of games from multiple countries and platforms over the last two decades.
- **Identify innovations:** We will define a "novel genre combination" as a new set of gameplay tags not seen in earlier games. Using our data, for each year we will list the unique tag combinations of games. When a combination appears that we have not seen before in prior years, we mark it as an innovation. We record the year and the country of the first game that had this new combination.
- **Trace diffusion:** For each identified genre innovation, we will track when games with the same combination appear in other countries. This builds a country-to-country timeline of adoption. We can visualize this as a network or map, where edges connect origin and adopter countries.
- **Analyze diffusion factors:** To understand why some pathways are faster or slower, we will analyze factors like (a) Distance: geographic distance between countries; (b) Language/culture: whether countries share a language or cultural ties; (c) Industry links: e.g. if the games share the same publisher or development partnerships across countries; (d) Platform overlap: if a genre innovation originates on a platform (like PC vs console) that is popular in certain regions; (e) Market size: the size of each country's game industry. We will use statistical models (e.g. survival analysis or regression) to test which factors predict the speed or order of adoption. For example, we might test if games in countries that share a language adopt an innovation faster than more distant countries.

- **Analysis tools:** We will use graph/network analysis software (such as Gephi or Python network libraries) to visualize diffusion paths, and statistical software (R or Python) to run regression models. All methods will be explained in clear steps, and we will ensure the definitions (like what counts as a “tag combination”) are transparent.

5. Expected Outcomes & Significance

We expect to find clear patterns in how game-genre innovations spread. For example, we might observe that a new game mechanic often originates in one or two countries (perhaps where game development is strong) and then reaches others. It could be that innovations move fastest along “invisible highways” like common languages (e.g. games developed in English-speaking countries appear quickly in other English-speaking markets) or global publisher networks.

We will produce maps and charts showing the origin and spread of several concrete genre innovations. For instance, we might show that a popular new hybrid genre first appeared in Country A in 2018 and then diffused to neighboring Country B in 2019, and to distant Country C in 2020. We will quantify how factors like distance or shared language correlate with diffusion speed.

Significance: This study will offer new insights into cultural and technological diffusion. Theoretically, it will expand diffusion models to creative digital products. Practically, it can help game developers and industry analysts by identifying how ideas travel – for example, informing indie studios where their innovative games might find receptive audiences. Policymakers interested in fostering creative industries can also learn whether building international networks accelerates innovation spread. In a rapidly growing global game market, understanding these “hidden pathways” can benefit anyone aiming to encourage or track video game innovation.

6. References

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